

WASP-84b

R-1179

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-84_210215 · created 2026-07-31T04:15:11+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459260.8068 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.125 +/- 0.0067
Transit depth (Rp/Rs)^2	1.56 +/- 0.17 [%]
Orbital Inclination (inc)	88.74 +/- 0.34
Airmass coefficient 1 (a1)	1.0172 +/- 0.0061
Airmass coefficient 2 (a2)	-0.0126 +/- 0.0066
Scatter in the residuals of the lightcurve fit is	0.6 %
Best Comparison Star	#3 - [365, 174]
Optimal Aperture	5.48
Optimal Annulus	21.81
Transit Duration (day)	0.1251 +/- 0.0079

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.125 ± 0.0067 vs 0.1295 (deviation 0.59σ)
Duration fit vs published	0.1251 d vs 0.1146 d (deviation 1.17σ)
Mid-time O–C	0.1 min
Depth SNR	16.4
Residual scatter	0.6 %

Light curve

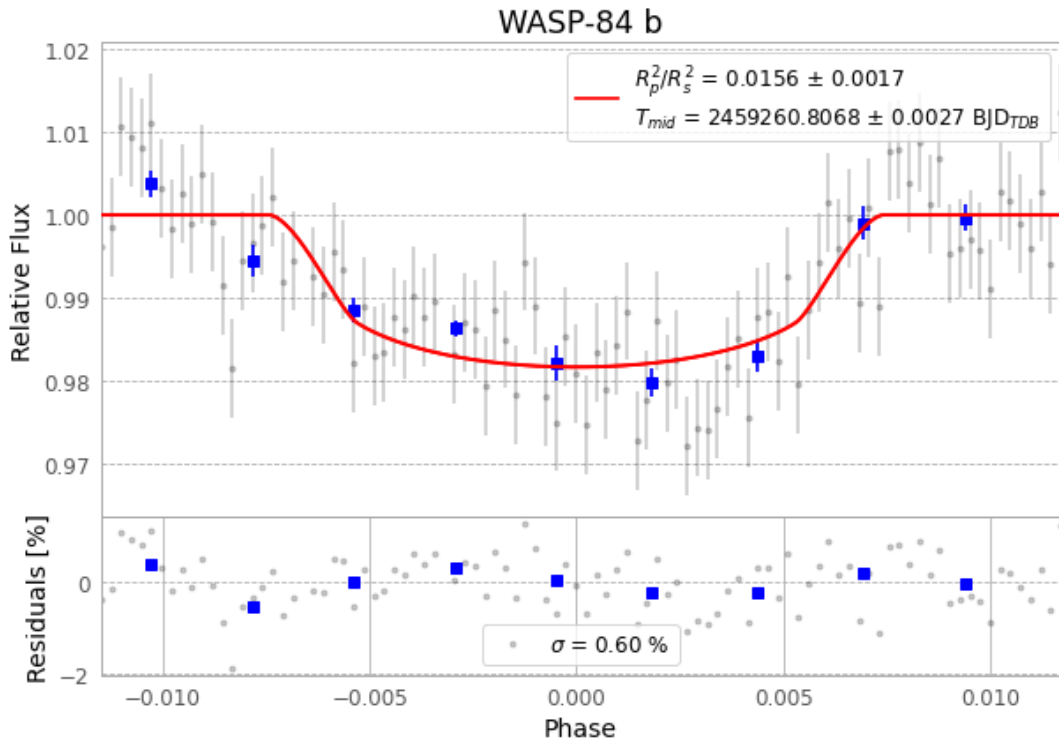


Figure 1 — FinalLightCurve_WASP-84 b_2021-02-14.png (SHA-256 3f3709bd43710264...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [415, 240], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bd878506e87e4ff9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

96 FITS frames (63 MB), WASP-84210215045112.FITS → WASP-84210215093612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-84-210215.log (16546e085ca3...), FinalLightCurve_WASP-84 b_2021-02-14.png (3f3709bd4371...), FinalLightCurve_WASP-84 b_2021-02-14.csv (3abf10765162...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 04:15Z.